

math111a hw4

jc — October 25, 2025

Definition: The centraliser of a subset

Let G be a group and $A \subseteq G$

The centraliser of A is the set $C_G(A) = \{g \in G : gag^{-1} = a \forall a \in A\} = \{g \in G : ga = ag \forall a \in A\}$

Definition: Normalizer of a group

Let G be a group and $A \subseteq G$

For $g \in G$ define $gAg^{-1} = \{gag^{-1} : a \in A\}$

The normaliser of A in G is defined as $N_G(A) = \{g \in G : gAg^{-1} = A\}$

Problem 1

In each part show that for the specified group G and subgroup A of G , $C_G(A) = A$ and $N_G(A) = G$

- a) $G = S_3$ and $A = \{1, (1\ 2\ 3), (1\ 3\ 2)\}$
- b) $G = D_8$ and $A = \{1, s, r^2, sr^2\}$
- c) $G = D_{10}$ and $A = \{1, r, r^2, r^3, r^4\}$

Solution: First we show $C_G(A) = A$ for each part

- a) Let $\sigma \in A$ then since A is cyclic and abelian, then $\sigma \in C_{S_3}(A)$

The other elements of S_3 not in A are $\{(1\ 2), (2\ 3), (1\ 3)\}$ (let's check if they commute with $(1\ 2\ 3)$)

We can check $(1\ 2\ 3)(1\ 2) = (1\ 3)$ and $(1\ 2)(1\ 2\ 3) = (2\ 3)$

We can check $(1\ 2\ 3)(2\ 3) = (1\ 2)$ and $(2\ 3)(1\ 2\ 3) = (1\ 3)$

We can check $(1\ 2\ 3)(1\ 3) = (2\ 3)$ and $(1\ 3)(1\ 2\ 3) = (1\ 2)$

Therefore $C_{S_3}(A)$ cannot have elements outside of $A \therefore \sigma \in C_{S_3}(A) \Rightarrow \sigma \in A$ and $C_{S_3}(A) = A$

- b) From the Cayley table for D_8 in homework 2, we can verify that all elements of A commute with each other.
Therefore $x \in A \Rightarrow x \in C_G(A)$

The other elements in D_8 are $\{sr, sr^3, r, r^3\}$ (let's see if they commute with s)

We can check $(sr)(s) = srs = ssr^3 = r^3 \neq r = (s)(sr)$

We can check $(sr^3)(s) = sr^3s = ssr = r \neq r^3 = (s)(sr^3)$

We can check $(r)(s) = sr^3 \neq (s)(r)$

We can check $(r^3)(s) = sr \neq (s)(r^3)$

Therefore $C_{D_8}(A)$ cannot have elements outside of $A \therefore x \in C_{D_8}(A) \Rightarrow x \in A$ and $C_{D_8}(A) = A$

- c) Since A is the cyclic group generated by r then A is abelian and $x \in A \Rightarrow x \in C_{D_{10}}(A)$

The other elements in D_{10} are $\{s, sr, sr^2, sr^3, sr^4\}$ (let's see if they commute with r)

Problem 2

Let H be a subgroup of group G .

- Show that $H \leq N_G(H)$. give an example to show this is not necessarily true if H is not a subgroup of G
- Show that $H \leq C_G(H)$ iff H is abelian

Solution:

- Since H is a subgroup, it is closed and inverses exist, therefore we only need to show that $H \subseteq N_G(H)$

Let $h \in H$, then we would like to show that $hHh^{-1} = H$

$\subseteq$ let an element $z \in hHh^{-1}$, then $z = hh'h^{-1}$ where $h' \in H$. Since H is closed, then $z \in H$

$\supseteq$ let an element $z \in H$ then we want to show that $z = hh'h^{-1}$ for some $h' \in H$. Since H is closed, then $hh'h^{-1} \in H$ and we can set $h' = h^{-1}zh \in H$ to obtain the desired result.

The claim follows.

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Say that H is only a subset of G say $H = \{(1\ 2), (1\ 3)\} \subseteq S_3 = G$

Then $(1\ 2)(1\ 2)(2\ 1) = () \notin H$ and $(1\ 2)(1\ 3)(2\ 1) = (2\ 3) \notin H$

If $(1\ 2) \in H$ is in $N_{S_3}(H)$ then it has to normalize something in H , but the above show it does not, therefore $x \in H \nRightarrow x \in N_{S_3}(H)$ in this case $\therefore H \not\leq N_{S_3}(H)$

- $\Rightarrow$ let $x \in H$ then $x \in C_G(H) \Rightarrow xh = hx \forall h \in H \Rightarrow xh = hx \forall h, x \in H$ (since x was chosen arbitrarily) which means that H is abelian

$\Leftarrow$ Let H be abelian then we want to show $H \subseteq C_G(H)$

Let $x \in H$ then $xh = hx \forall h, x \in H$ by assumption $\Rightarrow x \in C_G(H)$ by definition, and since H is closed and inverses exist then $H \leq C_G(H)$

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Problem 3

Find all generators of $\mathbb{Z}_{48}$

Solution:

1, 5, 7, 11, 13, 17, 19, 23, 25, 29, 31, 35, 37, 41, 43, 47

Since the generators of $\mathbb{Z}_n$ must be coprime to n , which come from the fact that if $H = \langle x \rangle$ and $|x| = n < \infty$ then x^a generates H iff $(a, n) = 1$. Here $x = 1$ with order $n = 48$

Sources:

David and Foote proposition 6.2, Section 2.3

Problem 4

What is the order of $\overline{30} \in \mathbb{Z}_{54}$? Write out all the elements and their orders in $\langle \overline{30} \rangle$

Solution:

The order is $54/(30, 54) = 54/6 = 9$ (by the proposition from class)

The elements of the subgroup generated by $\overline{30}$ are $\{\overline{0}, \overline{30}, \overline{6}, \overline{36}, \overline{12}, \overline{42}, \overline{18}, \overline{48}, \overline{24}\}$ with orders $\{1, 9, 9, 3, 9, 9, 3, 9, 9\}$ respectively, by the same proposition

Sources:

David and Foote proposition 5.2, Section 2.3

Problem 5

Find all the cyclic subgroups of D_8 . Find a proper subgroup D_8 that is not cyclic

Solution:

Note that $D_8 = \{1, r, r^2, r^3, s, sr, sr^2, sr^3\}$, the cyclic subgroups are

$$\langle 1 \rangle = \{1\}$$

$$\langle s \rangle = \{1, s\} \text{ since } s^2 = 1$$

$$\langle sr \rangle = \{1, sr\} \text{ since } sr sr = s r r^{-1} s = s^2 = 1$$

$$\langle sr^2 \rangle = \{1, sr^2\} \text{ since } sr^2 s s^2 = s r^2 r^{-1} s = 1$$

$$\langle sr^3 \rangle = \{1, sr^3\} \text{ by similar reasoning}$$

$$\langle r \rangle = \{1, r, r^2, r^3\}$$

$$\langle r^2 \rangle = \{1, r^2\}$$

$$\langle r^3 \rangle = \{1, r^3, r^2, r^1\} = \langle r \rangle$$

A subgroup that is not cyclic is $\langle s, r^2 \rangle = \{1, r^2, s, sr^2\}$ but each of $1, s, r^2, sr^2$ by itself cannot generate $\langle s, r^2 \rangle$ (see the subgroups above), this is a subgroup of D_8 since it is closed and a non-empty finite subset of D_8

Problem 6

Prove that if x is the only element of order 2 in a group, then x lies in the centre of the group

Solution:

Let $x \in G$ where G is a group and assume that $x^2 = 1$

Note that for all $g \in G$ the element gxg^{-1} also has order 2 since $(gxg^{-1})(gxg^{-1}) = gxg^{-1}gxg^{-1} = e$

By assumption, this means $(\forall g)$ that $x = gxg^{-1} \Rightarrow xg = gx$ i.e. that x is in the center of G

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Problem 7

- a) Prove that D_6 does not have a subgroup of order 4
 b) Let H be a subgroup of D_{2n} of odd order. Prove that every member of H is a rotation

Solution:

a) $D_6 = \{e, r, r^2, s, sr, sr^2\}$ so there are 6 choose 4 = $\frac{6!}{(6-4)!4!} = 15$ possible sets of D_6 with 4 elements

$\{e, r, r^2, s\} : r \cdot r^2 = r^3 \Rightarrow$ not closed

$\{e, r, r^2, sr\} : r \cdot r^2 = r^3 \Rightarrow$ not closed

$\{e, r, r^2, sr^2\} : r \cdot r^2 = r^3 \Rightarrow$ not closed

$\{e, r, s, sr\} : r \cdot s = rs \neq sr \Rightarrow$ not closed

$\{e, r, s, sr^2\} : s \cdot sr^2 = r^2 \Rightarrow$ not closed

$\{e, r, sr, sr^2\} : r \cdot sr = rr^{-1}s = s \Rightarrow$ not closed

$\{e, r^2, s, sr\} : r^2 \cdot r^2 = r^4 = r^1 = r \Rightarrow$ not closed

$\{e, r^2, s, sr^2\} : r^2 \cdot r^2 = r \Rightarrow$ not closed

$\{e, r^2, sr, sr^2\} : r^2 \cdot sr = r^2 \cdot r^{-1}s = rs \Rightarrow$ not closed

$\{e, s, sr, sr^2\} : s \cdot sr = r \Rightarrow$ not closed

$\{r, r^2, s, sr\} : \text{no identity}$

$\{r, r^2, s, sr^2\} : \text{no identity}$

$\{r, r^2, sr, sr^2\} : \text{no identity}$

$\{r, s, sr, sr^2\} : \text{no identity}$

$\{r^2, s, sr, sr^2\} : \text{no identity}$

Then D_6 does not have a subgroup of order 4 (also by lagrange theorem)

b)

Note that the order of H is finite, since it is a subgroup of a finite group.

If H is a finite group and $x \in H$, then the order of x divides the order of H . If such an x is a reflection, it must have order 2 but $2 \nmid |H|$ since $|H|$ is odd. Therefore, H can only contain rotations. To see that such an H exists, consider the subgroup $\langle r \rangle \leq D_{10}$ where $\langle r \rangle = \{e, r, r^2, r^3, r^4\}$

Sources:

David and Foote, Proposition 9, Section 3.2

<https://homepages.math.uic.edu/~groves/teaching/2008-9/330/09-330HW7Sols.pdf>

Definition: centre of a group

the centre of a group is defined as $Z(G) = \{g \in G : gx = xg \forall x \in G\}$

Note

$Z(G) = C_G(G)$ and is a subgroup of G

Definition: Centraliser of an element

Let G be a group and $a \in G$ fixed

The centraliser of a in G is the set $C(a) = \{g \in G : gag^{-1} = a\} = \{g \in G : ga = ag\}$

Problem 8

Let G be a group. Show that $Z(G) = \bigcap_{a \in G} C(a)$

Solution: show set equality

$\subseteq$

Let $g \in Z(G)$ then $ga = ag \forall a \in G$.

Fix any $a \in G$ then $g \in \{g \in G : ga = ag\} = C(a)$

Then $g \in \bigcap_{a \in G} C(a)$

$\supseteq$

Let $g \in \bigcap_{a \in G} C(a)$ then for any $a \in G$ we have $g \in C(a)$ i.e. that $ga = ag \forall a$

This implies that $g \in \{g \in G : ga = ag \forall a \in G\} = Z(G)$

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Problem 9

Let G be an abelian group and $H = \{x \in G : |x| \text{ is odd}\}$. Prove that H is a subgroup of G .

Solution:

Firstly, $1 \in H$ since $|1| = 1$ therefore $H \neq \emptyset$

Next, note that $\forall x \in H, |x^{-1}| = |x| \Rightarrow |x^{-1}| \text{ is odd} \Rightarrow x^{-1} \in H$

Next, let $a, b \in H$ and set the orders $m := |a|, n := |b|, r := \text{lcm}(m, n) \Rightarrow r = mk, r = nl$ for some integers k, l

Then $(ab)^r = a^r b^r = a^{mk} b^{nl} = (a^m)^k (a^n)^l = 1$ since G is abelian, therefore the order of ab is $r = \text{lcm}(m, n)$

Since m, n are odd then $\text{lcm}(m, n)$ is odd¹ and therefore ab has odd order i.e. $ab \in H$

Then since H is a subset of G , finite, closed, and each element of H has an inverse in H then the claim follows.

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1: since $ab = \gcd(a, b) \cdot \text{lcm}(a, b)$ (for positive integers a, b)

Sources:

<https://math.hawaii.edu/~allan/AdditionalGroupTheory.pdf>

https://en.wikipedia.org/wiki/Least_common_multiple